

A full-spark hypercyclic backward shift on ω : every infinite suborbit is total

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Abstract

We give a negative answer to the locally convex version of the non-total suborbit question highlighted by Ansari. On the Fréchet sequence space $\omega = \mathbb{K}^{\mathbb{N}_0}$, the unilateral backward shift has a hypercyclic vector x for which every infinite suborbit is total. Indeed, such vectors form a residual subset of ω . The proof intersects the residual set of hypercyclic vectors with countably many open dense nonvanishing conditions for finite Hankel minors.

1 The question and the answer

Throughout, $\mathbb{K}$ is either $\mathbb{R}$ or $\mathbb{C}$. A subset S of a locally convex space is *total* when $\overline{\text{span}}S$ is the whole space. Ansari proves that if T acts on an infinite-dimensional normed space and x is supercyclic, then some infinite suborbit $\{T^{n_k}x : k \geq 1\}$ is not total [1, Theorem 4.4]. The paper ends by inviting investigation of the same question on locally convex spaces:

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for some $k \geq 1$ and we have already seen that $\text{dist}(x, \text{span}\{Y_k, a_k\}) > 1$. Therefore, the assumption $\text{dist}(x, \overline{M}) < 1$ cannot be true and this shows that $x \notin \bigvee \{T^{n_k}x : k \geq 1\}$. $\square$

We need to mention that the authors in [6] have also answered the above-mentioned open question independently. The interested readers are invited to investigate Question 1 of [8] for operators on locally convex spaces.

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We show that the extension fails, in the strongest possible way, on the classical Fréchet space

$$\omega = \mathbb{K}^{\mathbb{N}_0}$$

with its product topology. Let $B: \omega \rightarrow \omega$ be the continuous unilateral backward shift

$$B(x_0, x_1, x_2, \dots) = (x_1, x_2, x_3, \dots).$$

Theorem 1 (Residual hereditary totality). *There is a residual set $\mathcal{R} \subset \omega$ such that every $x \in \mathcal{R}$ is hypercyclic for B and, for every infinite set $J \subset \mathbb{N}_0$,*

$$\overline{\text{span}}\{B^n x : n \in J\} = \omega.$$

Corollary 2. *The locally convex extension of the non-total-suborbit assertion is false. More explicitly, there are an infinite-dimensional locally convex space X , a continuous linear operator T on X , and a supercyclic vector x such that*

$$\overline{\text{span}}\{T^{n_k}x : k \geq 1\} = X$$

for every strictly increasing sequence (n_k) of positive integers.

The point is that a single Baire-generic sequence can simultaneously be hypercyclic and have every finite row minor of its infinite Hankel matrix nonzero.

2 Two residual sets

We first record hypercyclicity in a form convenient for the proof.

Lemma 3. *The backward shift B is topologically mixing on ω . Consequently, its set $HC(B)$ of hypercyclic vectors is a dense G_δ subset of ω .*

Proof. Let U, V be nonempty basic open cylinders. Membership in each cylinder constrains only finitely many coordinates. For every sufficiently large n , the coordinates constrained by U and those of the form $n + j$ needed to ensure $B^n z \in V$ are disjoint. We may therefore choose z satisfying both sets of constraints. Thus $B^n(U) \cap V \neq \emptyset$ for all large n , proving mixing.

For completeness, let $(V_r)_{r \geq 1}$ be a countable base for ω and put

$$G_r = \bigcup_{n \geq 0} B^{-n}(V_r).$$

Each G_r is open and dense by transitivity. Since ω is a complete metrizable space, Baire's theorem gives that $\bigcap_r G_r$ is dense G_δ . A vector belongs to this intersection exactly when its orbit meets every basic open set, so $\bigcap_r G_r = HC(B)$. $\square$

For $d \geq 0$ and $0 \leq n_0 < \dots < n_d$, define the polynomial

$$\Delta_{d,\mathbf{n}}(x) = \det[x_{n_i+j}]_{i,j=0}^d, \quad x = (x_k)_{k \geq 0} \in \omega.$$

Lemma 4 (Full-spark Hankel sequences). *For each d and $\mathbf{n} = (n_0, \dots, n_d)$, the set*

$$\mathcal{U}_{d,\mathbf{n}} = \{x \in \omega : \Delta_{d,\mathbf{n}}(x) \neq 0\}$$

is open and dense. Hence

$$\mathcal{F} = \bigcap_{d \geq 0} \bigcap_{0 \leq n_0 < \dots < n_d} \mathcal{U}_{d,\mathbf{n}}$$

is residual in ω .

Proof. Continuity of the finitely supported coordinate polynomial $\Delta_{d,\mathbf{n}}$ gives openness. To prove density, it is enough to see that this polynomial is not identically zero. Fix any real $q > 1$ and evaluate it at the sequence $u_k = q^{k^2}$. Since

$$u_{n_i+j} = q^{(n_i+j)^2} = q^{n_i^2} (q^{2n_i})^j q^{j^2},$$

we obtain

$$\begin{aligned}\Delta_{d,\mathbf{n}}(u) &= \left(\prod_{i=0}^d q^{n_i^2} \right) \left(\prod_{j=0}^d q^{j^2} \right) \det [(q^{2n_i})^j]_{i,j=0}^d \\ &= \left(\prod_{i=0}^d q^{n_i^2} \right) \left(\prod_{j=0}^d q^{j^2} \right) \prod_{0 \leq i < k \leq d} (q^{2n_k} - q^{2n_i}),\end{aligned}$$

which is nonzero. A nonzero real or complex polynomial cannot vanish on a nonempty open subset of a finite-dimensional space. Thus, inside any basic cylinder of ω , the finitely many coordinates occurring in $\Delta_{d,\mathbf{n}}$ can be chosen so that the determinant is nonzero. This proves density. There are only countably many pairs $(d, \mathbf{n})$, so the last assertion follows from Baire's theorem. $\square$

3 Every infinite suborbit is total

We use the elementary finite-projection characterization of density in a product sequence space. Write

$$\pi_d: \omega \longrightarrow \mathbb{K}^{d+1}, \quad \pi_d(x) = (x_0, \dots, x_d).$$

Lemma 5. *If a linear subspace $M \subset \omega$ satisfies $\pi_d(M) = \mathbb{K}^{d+1}$ for every $d \geq 0$, then M is dense in ω .*

Proof. Every nonempty basic open set in the product topology constrains only finitely many coordinates, all contained in $\{0, \dots, d\}$ for some d . Surjectivity of $\pi_d|_M$ lets an element of M match any prescribed values on this whole initial block, and hence meet the basic open set. $\square$

Proof of Theorem 1. Let

$$\mathcal{R} = HC(B) \cap \mathcal{F}.$$

By Lemmas 3 and 4, this is a residual subset of the Baire space ω .

Fix $x \in \mathcal{R}$, an infinite $J \subset \mathbb{N}_0$, and $d \geq 0$. Choose distinct $n_0 < \dots < n_d$ from J . The coordinate vectors

$$\pi_d(B^{n_i}x) = (x_{n_i}, x_{n_i+1}, \dots, x_{n_i+d}), \quad 0 \leq i \leq d,$$

form the rows of the matrix whose determinant is $\Delta_{d,\mathbf{n}}(x)$. This determinant is nonzero because $x \in \mathcal{F}$. The vectors therefore form a basis of $\mathbb{K}^{d+1}$, so

$$\pi_d(\text{span}\{B^n x : n \in J\}) = \mathbb{K}^{d+1}.$$

As this holds for every d , Lemma 5 shows that the span is dense in ω . $\square$

Proof of Corollary 2. Choose any $x \in \mathcal{R}$ and take $X = \omega$, $T = B$. The space is infinite-dimensional, Hausdorff, locally convex and Fréchet. The vector x is hypercyclic, hence supercyclic because the projective orbit contains the ordinary orbit. For any strictly increasing sequence of positive integers, its range is an infinite subset $J \subset \mathbb{N}_0$, so Theorem 1 gives the asserted equality. $\square$

4 Relation to earlier work

Bernal-González and Bonilla study total and non-total suborbits for hypercyclic operators and obtain affirmative normed-space results as well as constructions on Fréchet spaces admitting a continuous norm [2]. The space ω admits no continuous norm, and the present result exhibits the opposite behavior relevant to Ansari's question: not merely one selected suborbit, but *every* infinite suborbit of the chosen vector is total. This does not conflict with the normed-space theorem; it isolates the product topology as the source of the failure.

Novelty note. A bounded search of arXiv and the web on 21 July 2026 recovered the two works cited below but no statement of the residual full-spark construction above. This is evidence, not a definitive literature certification.

References

- [1] M. Ansari, *Three results in linear dynamics*, arXiv:2310.14099 (2023), <https://arxiv.org/abs/2310.14099>.
- [2] L. Bernal-González and A. Bonilla, *Total and non-total suborbits for hypercyclic operators*, Rev. Real Acad. Cienc. Exactas Fís. Nat. Ser. A-Mat. 117, 17 (2023), <https://link.springer.com/article/10.1007/s13398-022-01351-0>.